Data Preparation

Date	29 October 2026
Team ID	Not provided — confirm from MySkillWallet/college
Project Name	EV Market Intelligence Dashboard
Maximum Marks	5 Marks

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1: Identifying and Handling Null Values

Identify the NULL or missing values present in the Snowflake dataset, analyze the affected columns, determine the appropriate approach for handling them based on the data and business requirements, apply the required query, and validate the cleaned data to ensure it is ready for further analysis and visualization.

[Give Codes with Output]

Activity 2: Duplicate Record Handling

Identify and analyze duplicate records in the Snowflake dataset, determine whether the duplicates are genuine or repeated entries, define the appropriate criteria for identifying duplicates, remove or retain records based on the project requirements, and validate the dataset after treatment to ensure that duplicate records do not affect the accuracy of further analysis and visualizations.

[Give Codes with Output]

Activity 3: Data Cleaning and Standardization

Identify and correct inconsistencies, invalid entries, formatting issues, and data quality problems in the Snowflake dataset, standardize values and formats across relevant columns, apply the required SQL-based cleaning and transformation steps, and validate the cleaned and standardized dataset to ensure consistency, accuracy, and readiness for further analysis and visualization.

[Give Codes with Output]

Activity 4: Data Transformation and Validation

Apply the required data transformations to prepare the cleaned data for analysis and modelling. Create necessary derived fields, apply business rules, filter or transform records where required, and validate the prepared data and other relevant quality checks before using it for the CURATED and MART layers.

[Give Codes with Output]

Dataset-Specific Data Preparation Record

Source file: ev_market_test.csv. Shape: 100,000 rows × 37 columns. This workflow uses Tableau Public and CSV; the template's Snowflake RAW/STAGING/CURATED/MART layers are not part of the actual implementation.

Activity 1: Identifying and Handling Null Values

- resale_value_pct_3yr: 3,057 null values.
- battery_cycle_life: 2,028 null values.
- customer_rating: 1,931 null values.
- infotainment_screen_inch: 1,506 null values.
- annual_sales_units: 989 null values.

The file contains missing values in several fields. For aggregate charts, AVG calculations naturally ignore null values; record counts for battery type use Vehicle Id, which has no nulls in the inspected file. Do not replace missing customer ratings, resale values, or sales units with zero unless the project requirements explicitly justify it.

Validation evidence: add a screenshot of the data pane/profile or a worksheet showing null counts. Tableau Public web authoring may not provide full data-quality profiling, so a CSV check can be documented separately.

Activity 2: Duplicate Record Handling

Full-row duplicates found in the supplied CSV: 0. Duplicate vehicle_id values: 0. No duplicate removal was required based on these two checks. Keep the source unchanged and record the validation output.

Validation check (Python/pandas used for this report): df.duplicated().sum() and df['vehicle_id'].duplicated().sum(). These are data-validation checks, not SQL executed in Snowflake.

Activity 3: Data Cleaning and Standardization

The inspected CSV has 37 columns with numeric measures represented as numeric types and descriptive attributes such as category, manufacturer, battery_type, market_region, and manufacturing_country. Check category spelling/whitespace and country labels for consistency before interpreting groups. No unverified value replacement has been applied.

Recommended Tableau checks: inspect field data types; verify geographic role for Manufacturing Country; check nulls on each measure; use consistent number formats and units; ensure sales and CO_2 measures are not accidentally interpreted as counts.

Activity 4: Data Transformation and Validation

The nine views use the following aggregations: SUM(Annual Sales Units), COUNT(Vehicle Id), AVG(Range Km), AVG(Customer Rating), AVG(CO2 Saved Tons Per Year), plus Model Year and Battery Capacity Kwh in trend/scatter views. The top-manufacturer chart uses a Top 10 filter. Validate each worksheet's aggregation, sort order, filter and axis title against the CSV.

Important: This document describes the checks actually computed against the supplied CSV. It does not claim that Snowflake SQL, Snowflake tables, or a Snowflake warehouse were used.